

Fernando d'Ávila Lins Bezerra Cavalcanti Filho

João Pessoa, Brazil | fernando.d.avila.filho@gmail.com | +55 83 99602-2155
GitHub | LinkedIn

Summary

Computer Science student at UFPB and software developer with professional experience in full-stack development and hands-on experience in Artificial Intelligence, Machine Learning, NLP, and LLMs. Experienced in research and development projects involving RAG, natural language processing, computer vision, data analysis, and AI-powered applications. Skilled in Python, TypeScript, Node.js, React, REST APIs, PostgreSQL, and Docker, with a strong interest in turning research and real-world problems into functional AI systems.

Technical Skills

AI / Machine Learning: Python, Machine Learning, Deep Learning, LLMs, RAG, NLP, Computer Vision, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy

AI Engineering: Prompt Engineering, Embeddings, Information Retrieval, LLM Integration, AI Agents, MCP

Software Engineering: TypeScript, JavaScript, Node.js, React, Next.js, REST APIs, Git, Docker

Data: PostgreSQL, SQL, Prisma, Data Processing, Data Pipelines

Professional Experience

Acutis Tecnologia

Full-Stack Development Intern

May 2024 – Present

João Pessoa, Brazil

- Develop and maintain web applications using JavaScript, TypeScript, Next.js, Node.js, APIs, and databases.
- Build frontend and backend features, system integrations, and contribute to debugging, testing, and application maintenance.
- Contribute to solutions involving OCR and automated document processing.
- Developed a flashcard application designed for medical students.
- Translate business requirements and user needs into functional software solutions.

OpenDoor App

Software Development Intern

Jun 2025 – Present

Remote

- Develop a full-stack international home-swapping platform using Next.js, Node.js, TypeScript, PostgreSQL, and Prisma.
- Implement and maintain REST APIs, business logic, database integrations, and frontend features.
- Work across the product stack, considering how database and backend decisions affect APIs, frontend behavior, and the end-user experience.
- Investigate and resolve bugs spanning multiple layers of the application.

Colégio e Curso Evolução

Education & Technology Intern

Jan 2024 – Jan 2025

João Pessoa, Brazil

- Developed technology solutions for educational and administrative processes.
- Built a Python solution to automate a manual teacher scheduling and allocation workflow.
- Gathered requirements and business rules directly from stakeholders and translated them into software solutions.

Artificial Intelligence & Research

TAIL – Technology and Artificial Intelligence League

AI Research Member

2024 – Present

UFPB

- Conduct research and develop projects involving Artificial Intelligence, NLP, and Large Language Models.
- Research zero-shot product title normalization using LLMs, comparing multiple models against traditional Machine Learning baselines.
- Evaluate models using metrics such as F1 score, prediction consistency, and stability across multiple executions.
- Develop and experiment with RAG-based applications and LLM-powered information retrieval workflows.

Selected AI Projects

Multiple Sclerosis Lesion Segmentation from MRI

- Developing an AI-based healthcare project for automatic segmentation of Multiple Sclerosis lesions in MRI scans.
- Exploring Computer Vision and Deep Learning techniques for lesion detection and quantification.

- Developing a workflow for 3D visualization of segmentation results to support analysis of lesion progression.

Zero-Shot Product Title Normalization with LLMs

- Researched the use of Large Language Models for zero-shot normalization of e-commerce product titles.
- Compared multiple LLMs and traditional baselines, analyzing performance, consistency, and inference stability.
- Built a Python experimental pipeline for data processing, model execution, and evaluation.

AI-Powered Football Narrator

- Developed a multimodal pipeline for automatically generating football commentary from match frames.
- Combined vision-language models for visual understanding and text generation with text-to-speech technology.

Sentiment Analysis with Neural Networks

- Developed and evaluated RNN, LSTM, and GRU models for sentiment classification on textual data.
- Implemented preprocessing, training, and evaluation pipelines using Python and Deep Learning frameworks.

Education

Federal University of Paraíba – UFPB

B.Sc. in Computer Science

2022 – Present

João Pessoa, Brazil

- Areas of interest: Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, and Software Engineering.

Additional Experience

Fórmula E-motion – UFPB

2025 – Present

Member of a multidisciplinary engineering team focused on developing technology for an electric vehicle.

Teaching Initiation Program – UFPB

2023

Tutor for Vector Calculus and Analytic Geometry. Recognized with UFPB’s Teaching Initiation Award for contributions to student learning.

Estrutura Negócios – Extension Project

2022 – 2023

Taught Lean Startup, MVP validation, and business modeling to entrepreneurs, and contributed to the development of a management system using JavaScript and TypeScript.

Languages

Portuguese: Native **English:** Professional Working Proficiency